The documentation and process conversion measures necessary to comply with this revision shall be completed by 28 January 1996

INCH-POUND
MIL-S-19500/291E
28 July 1994
SUPERSEDING
MIL-S-19500/291D
2 September 1991

MILITARY SPECIFICATION
SEMICONDUCTOR DEVICE, TRANSISTOR, PNP,
SILICON, SWITCHING,
TYPES 2N2906A, 2N2907A, 2N2906AUA
2N2907AUA, 2N2906AUB, AND 2N2907AUB
JAN, JANTX, JANTXV, JANS, JANHC, AND JANKC

This Specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for PNP, silicon, switching transistors. Four levels of product assurance are provided for each device type as specified in MIL-S-19500. Two levels of product assurance are provided for die.

1.2 Physical dimensions. See figures 1 and 2 (similar to a TO-18), figures 3 and 4 (surface mount case outlines UA and UB), and figure 5 (JANC).

1.3 Maximum ratings.

Types	P_T 1/ T_A = +25°C	P_T 2/,3/ T_C = +25°C	I_C	V_{CBO}	V_{CEO}	V_{EBO}	T_J and T_{STG}
	W	W	mA dc	V dc	V dc	V dc	°C
2N2906A,UA,UB	0.4 1/	1.8 2/ 3/	600	60	60	5	-65 to +200
2N2907A,UA,UB	0.4 1/	1.8 2/ 3/	600	60	60	5	-65 to +200

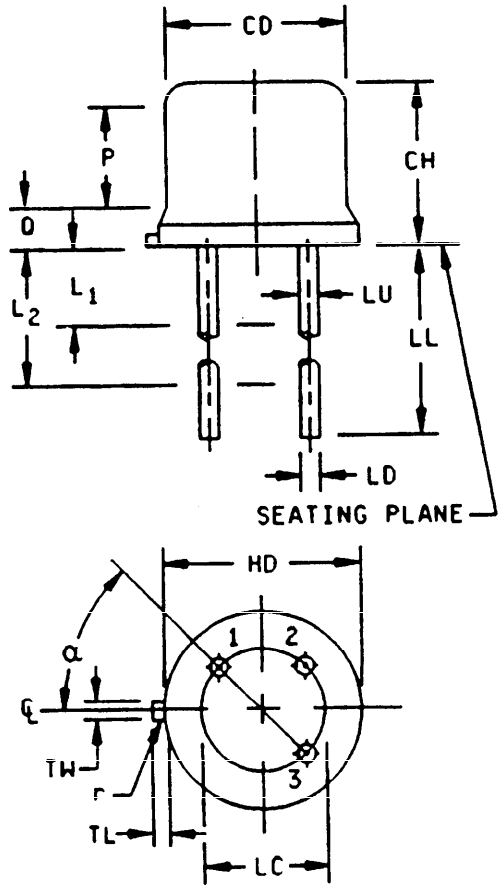
1/ Derate linearly 2.28 mW/°C for T_A > +25°C.
2/ Derate linearly 10.3 mW/°C for T_C > +25°C (Nonsurface mount devices).
3/ For UA and UB surface mount case outlines, P_T = 1.16 W; derate linearly 6.6 mW/°C for T_C > +25°C.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Research Laboratory Electronics and Power Sources Directorate, ATTN: AMSRL-EP-RD, Ft. Monmouth, NJ 07703-5601 by using the Standardization Document Proposal (DD Form 1426) appearing at the end of this document or by letter.

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Symbol	Inches		Millimeters		Note
	Min	Max	Min	Max	
CH	.170	.210	.8255	.8509	
LC	.100 TP		2.54 TP		6
LD	.016	.021	0.41	0.53	7,8
LU	.016	.019	0.41	0.48	7,8
HD	.209	.230	5.31	5.84	
CD	.178	.195	4.52	4.95	
Q	---	.030	---	0.76	5
TH	.036	.046	0.91	1.17	3
TL	.028	.048	0.71	1.22	3,4
LL	.500	.750	12.70	19.05	7,8
L1	---	.050	---	1.27	7,8
L2	.250	---	6.35	---	7,8
r	---	.010	---	0.25	10
α	45° TP		45° TP		6

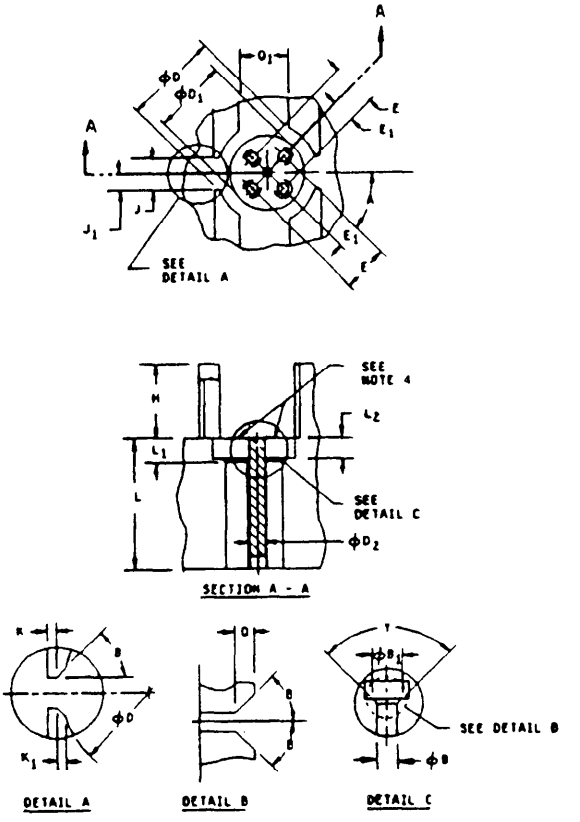


NOTES:
1. Dimension are in inches.
2. Metric equivalents are given for general information only.
3. Beyond r (radius) maximum, TH shall be held for a minimum length of .011 (0.28 mm).
4. Dimension TL measured from maximum HD.
5. Body contour optional within zone defined by HD, CD, and Q.
6. Leads at gauge plane .054 +.001 -.000 inch (1.37 +0.03 -0.00 mm) below seating plane shall be within .007 inch (0.18 mm) radius of true position (TP) at maximum material condition (MMC) relative to tab at MMC. The device may be measured by direct methods or by the gauge and gauging procedure shown in figure 2.
7. Dimension LU applies between L1 and L2. Dimension LD applies between L2 and LL minimum. Diameter is uncontrolled in L1 and beyond LL minimum.
8. All three leads.
9. The collector shall be internally connected to the case.
10. Dimension r (radius) applies to both inside corners of tab.

FIGURE 1. Physical dimensions (similar to TO-18).

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Ltr	Dimensions				Notes
	Millimeters		Inches		
	Min	Max	Min	Max	
ϕB	.8255	.8509	.0325	.0335	5
$\phi B1$	1.09 Nom		.040 Nom		5
ϕD	5.867	5.880	.2310	.2315	
$\phi D1$	4.04	4.09	.159	.161	
$\phi D2$	1.02 Nom		.040 Nom		6
E	2.527	2.553	.0995	.1005	
E1	1.257	1.283	.0495	.0505	
H	3.68	3.94	.145	.155	
J	1.194	1.207	.0470	.0475	
J1	.597	.622	.0235	.0245	
K	.229	.279	.009	.011	
K1	.127 Nom		.005 Nom		
L	9.45	9.60	.372	.378	
L1	1.37	1.40	.054	.055	
L2	1.09 Nom		.043 Nom		
Q	1.02 Nom		.040 Nom		
Q1	3.12	3.23	.123	.127	
α	---	---	44.90°	45.10°	
B	---		45° Nom		
Y	---		90° Nom		

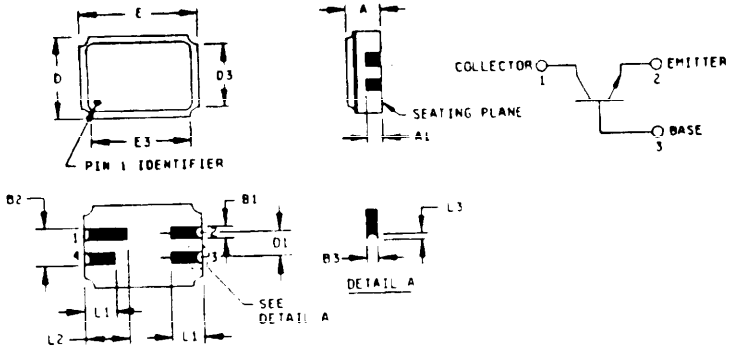


NOTES:

1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. The following gauging procedures shall be used: The device being measured shall be inserted until its seating plane is .125 inch (3.18 mm) +.010 inch (0.254 mm) from the seating surface of the gauge. A force of 8 ±.5 ounces shall then be applied parallel and symmetrical to the device's cylindrical axis. When examined visually after the force application (the force need not be removed) the seating plane of the device shall be seated against the gauge. The use of a pin straightener prior to insertion in the gauge is permissible. A spacer may be used to obtain the .125 inch (3.18 mm) distance from the gauge seat prior to force application.
4. These surfaces to be parallel and in same plane within ±.001 inch (0.025 mm).
5. Four holes.
6. Pressed in.

FIGURE 2. Gauge for lead and tab location.

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Dimensions					
Symbol	Inches		Millimeters		Notes
	Min	Max	Min	Max	
A	.061	.075	1.55	1.90	3
A1	.029	.041	0.74	1.04	
B1	.022	.028	0.56	0.71	
B2	.075 REF		1.91 REF		
B3	.006	.022	0.15	0.56	5
D	.145	.155	3.68	3.93	
D1	.045	.055	1.14	1.39	
D3	---	.155	---	3.93	
E	.215	.225	5.46	5.71	
E3	---	.225	---	5.71	
L1	.032	.048	0.81	1.22	
L2	.072	.088	1.83	2.23	
L3	.003	.007	0.08	0.18	5

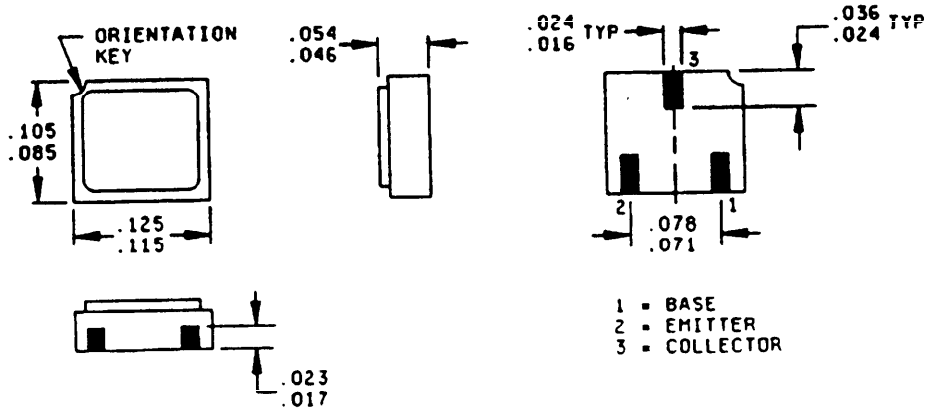
FIGURE 3. Physical dimensions, surface mount (UA version).

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NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Dimension "A" controls the overall package thickness. When a window lid is used, dimension "A" must increase by a minimum of .010 inch (0.254 mm) and a maximum of .040 inch (1.020 mm).
4. The corner shape (square, notch, radius, etc.) may vary at the manufacturer's option, from that shown on the drawing.
5. Dimensions "B3" minimum and "L3" minimum and the appropriately castellation length define an unobstructed three-dimensional space traversing all of the ceramic layers in which a castellation was designed. (Castellations are required on bottom two layers, optional on top ceramic layer.) Dimension "B3" maximum and "L3" maximum define the maximum width and depth of the castellation at any point on its surface. Measurement of these dimensions may be made prior to solder dipping.

FIGURE 3 Physical dimensions, surface mount (UA version) - Continued.

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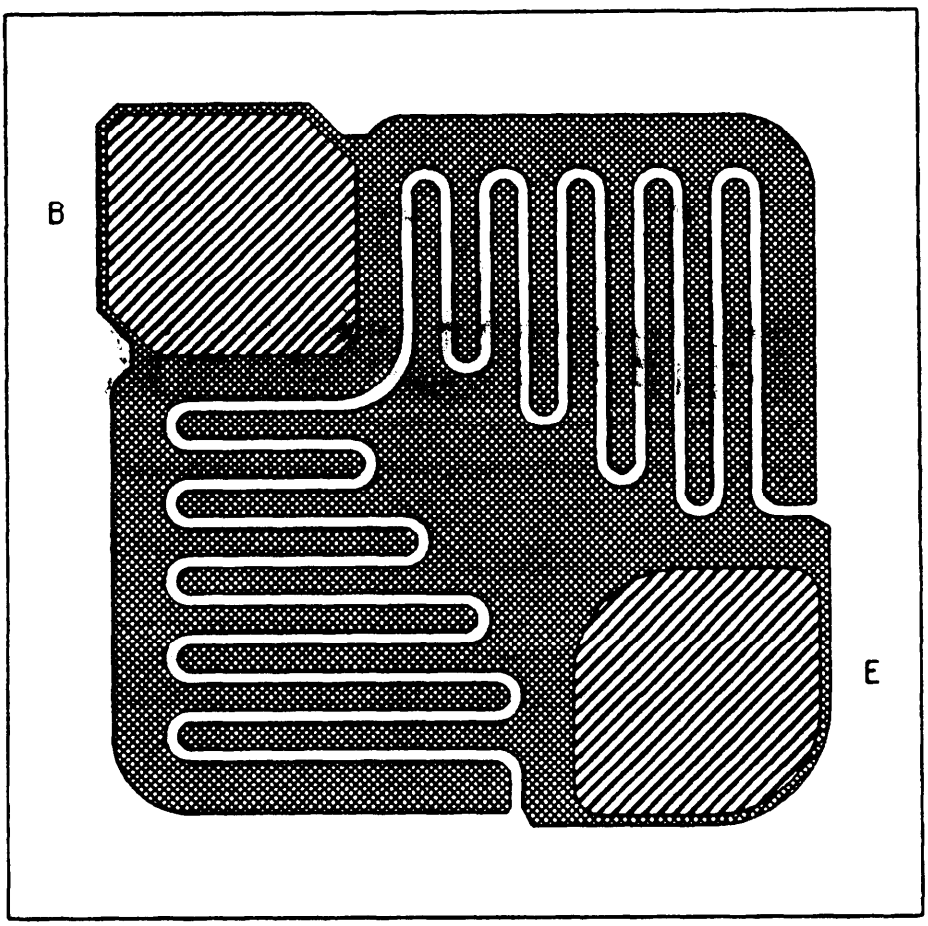


Inches	mm
.016	0.41
.017	0.43
.023	0.58
.024	0.61
.036	0.81
.046	0.97
.054	1.22
.071	1.81
.078	2.01
.085	2.41
.105	2.67
.115	2.82
.125	3.18

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.

FIGURE 4. Physical dimensions, surface mount (UB version).

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Physical characteristics:

Die size:	0.021 by 0.021 inches square, ±0.001.
Die thickness:	0.0090 ±0.0018 inches.
Base:	0.00440 square.
Emitter:	0.00445 square.
Back metal:	Gold 28 KÅ +3 KÅ, -1 KÅ.
Top metal:	Aluminum 23 KÅ ±2.3 KÅ.
Back side:	Collector.

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified tolerance is ±0.00100 inch (0.00254 mm).

FIGURE 5. JANH(and JANK((A-version) die dimensions.

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1.4 Primary electrical characteristics at T_A = +25°C.

	h_{FE} at V_{CE} = 10 V dc									
	h_{FE1} I_C = 0.1 mA dc		h_{FE2} I_C = 1.0 mA dc		h_{FE3} I_C = 10 mA dc		h_{FE4} 1/ I_C = 150 mA dc		h_{FE5} 1/ I_C = 500 mA dc	
	2N2906A UA, UB	2N2907A UA, UB	2N2906A UA, UB	2N2907A UA, UB	2N2906A UA, UB	2N2907A UA, UB	2N2906A UA, UB	2N2907A UA, UB	2N2906A UA, UB	2N2907A UA, UB
Min	40	75	40	100	40	100	40	100	40	50
Max			175	450			120	300		

Types	Limit	/hfe/ f = 100 MHz V_{CE} = 20 V dc I_C = 20 mA dc	Cobo 100 kHz ≤ f ≤ 1 MHz V_{CB} = 10 V dc I_E = 0	Switching (saturated)	
				ton See fig. 6	toff See fig. 7
			pF	ns	ns
2N2906A, 2N2907A UA, UB	Min	2.5			
	Max		8	45	300

Types	Limits	$V_{CE(sat)1}$ 1/ I_C = 150 mA dc I_B = 15 mA dc	$V_{CE(sat)2}$ 1/ I_C = 500 mA dc I_B = 50 mA dc	$V_{BE(sat)1}$ 1/ I_C = 150 mA dc I_B = 15 mA dc	$V_{BE(sat)2}$ 1/ I_C = 500 mA dc I_B = 50 mA dc
		V dc	V dc	V dc	V dc
2N2906A, 2N2907A UA, UB	Min				
	Max	0.4	1.6	1.3	2.6

1/ Pulsed see 4.5.1.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATION

MILITARY

MIL-S-19500 - Semiconductor Devices, General Specification for.

STANDARD

MILITARY

MIL-STD-750 - Test Methods for Semiconductor Devices.

(Unless otherwise indicated, copies of federal and military specifications, standards and handbooks are available from the standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein the text of this document take precedence. Nothing in this document, however, supersede applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 Associated detail specification. The individual item requirements shall be in accordance with MIL-S-19500, and as specified herein.

3.2 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-S-19500.

UA, UB, - - - - - - - Surface mount case outlines (see figures 3 and 4).

3.3 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in figures 1, 2, 3, 4, and 5.

3.3.1 Lead finish. Lead finish shall be solderable as defined in Mil-S-19500.

3.4 Marking. Marking shall be in accordance with MIL-S-19500.

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection shall be in accordance with MIL-S-19500, and as specified herein.

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-S-19500, and as specified herein.

4.3 Screening (JANS, JANTX, and JANTXV levels only). Screening shall be in accordance with MIL-S-19500 (table II), and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screen (see table II of MIL-S-19500)	Measurement	
	JANS level	JANTX and JANTXV levels
3c	Thermal impedance (see 4.3.3)	Thermal impedance (see 4.3.3)
7	hermetic seal	hermetic seal
9	I_{CBO1}, h_{FE2}	Not applicable
10	48 hours minimum	48 hours minimum
11	I_{CBO1}; I_{CES}; h_{FE2}; ΔI_{CBO1} = 100% of initial value or 5 nA dc, whichever is greater. Δh_{FE2} = ±15%	I_{CBO1}, h_{FE4}
12	See 4.3.2 240 hours minimum	See 4.3.2 80 hours minimum
13	Subgroups 2 and 3 of table I herein; ΔI_{CBO1} = 100% of initial value or 5 nA dc, whichever is greater; Δh_{FE2} = ±15%	Subgroup 2 of table I herein; ΔI_{CBO1} = 100% of initial value or 5 nA dc, whichever is greater; Δh_{FE4} = ±15%

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4.3.1 Screening (JANHC and JANKC). Screening of JANHC and JANKC die shall be in accordance with MIL-S-19500, appendix H. As a minimum, die shall be 100 percent probed to insure the assembled chips will meet the requirements of group A, subgroup 2.

4.3.2 Power burn-in conditions. Power burn-in conditions are as follows: T_A = Room ambient as defined in the general requirements of MIL-STD-750, paragraph 4.5;

JANS level (all device types) - - - - - - V_{CB} = 10 Vdc, P_T = 400 mW.
JANTX and JANTXV levels
2N2906A, UA; 2N2907A, UA - - - - - V_{CB} = 30 Vdc, P_T = 400 mW.
2N2906AUB, 2N2907AUB - - - - - V_{CB} = 30 Vdc, P_T = 200 mW.
NOTE: No heat sink or forced air cooling on the devices shall be permitted.

4.3.3 Thermal impedance ($Z_{\theta JX}$ measurements) (for qualification only). The $Z_{\theta JX}$ measurements shall be performed in accordance with MIL-STD-750, Method 3131.

a. I_M measurement current - - - - - - - - - - - - 5 mA.
b. I_H forward heating current - - - - - - - - - 200 mA (min).
c. t_H heating time - - - - - - - - - - - - - - - 25 - 30 ms.
d. t_{md} measurement delay time - - - - - - - - - 60 µs max.
e. V_{CE} collector-emitter voltage - - - - - - - 10 V dc

The maximum limit for $Z_{\theta JX}$ under these test conditions are $Z_{\theta JX}$ (max) = 72°C/W.

4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-S-19500, and as specified herein.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-S-19500, and table I herein.

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in table IVa (JANS) of MIL-S-19500. See 4.4.2.2 for JAN, JANTX, and JANTXV group B subgroup testing. Electrical measurements (end-points) and delta requirements shall be in accordance with the applicable steps of table II herein.

4.4.2.1 Group B inspection, table IVa (JANS) of MIL-S-19500.

Subgroup	Method	Condition
B3	1037	V_{CB} = 10 V dc; P_T = 400 mW at T_A = +25° ±3°C; t_{on} = t_{off} = 3 minutes minimum for 2,000 cycles. No heat sink or forced-air cooling on devices shall be permitted.
B5	1027	V_{CB} = 10 V dc; P_T = 400 mW at T_A = 100°C for 96 hours, or T_A = +125°C ±25°C for 96 hours with P_T adjusted according to the chosen T_A to give and average T_J = +275°C.

4.4.2.2 Group B inspection, (JAN, JANTX, and JANTXV).

Subgroup	Method	Condition
B1	1039	HTRB, test condition A, T_A = 150°C, V_{CB} = 50 V dc, 340 hours, 22 pcs, Acc = 0.
B2	1039	Steady-state life: Test condition B, 1,000 hours, V_{CB} = 10 V dc, P_T = 400 mW; P_T = 200 mW for UB; at T_A = +25° ±5°C, 45 devices, Acc = 0. No heat sink or forced-air cooling on the devices shall be permitted.
B2	2037	Bond strength, 11 wires, Acc = 0.

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4.4.2.3 Group B sample selection. Samples selected from group B inspection shall meet all of the following requirements:

a. Samples shall be selected from one lot every three months from a die fabrication line that the manufacturer chooses to "certify" for the die design used in products manufactured to this specification.

b. Must be chosen from an inspection lot that has been submitted to and passed group A quality conformance inspection.

4.4.2.4 Product acceptable for delivery. The product shall be acceptable for delivery only after the successful completion of all group B testing and shall be comprised of die meeting the following requirements:

a. Manufactured on the same certified die fabrication line as the sample selected for 4.4.2.3.

b. The die lot was manufactured (using start or completion dates, as the manufacturer selects to consistantly use) within three months following the lot used for sample selection in paragraph 4.4.2.3. Group B shall not be required more than once for any single die lot.

4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table V of MIL-S-19500. Electrical measurements (end points) and delta requirements shall be in accordance with the applicable steps of table II herein.

4.4.3.1 Group C inspection, table V (JANS) of MIL-S-19500.

Subgroup	Method	Condition
C2	2036	Test condition E.
C6	1026	conditions: 1,000 hours at V_{CB} = 10 V dc; P_T = 400 mW; P_T = 200 mW (for UB) at T_A = +25° ±3°C. No heat sink or forced-air cooling on device shall be permitted.

4.4.3.2 Group C inspection, JAN, JANTX, and JANTXV. Group C inspection (package related tests) shall be as follows and shall include those package or case-related tests which are performed periodically. Group C test on each package family shall be performed on devices from each six months production of devices (based on inspection lot identification codes) for each assembly location and die attach method. The sample size for each of these subgroups is: 22 devices, c = 0.

Subgroup	Method	Condition
C2	1051	Test condition C.
C2	2036	Test condition E.
C6	---	Not applicable.

4.4.3.3 Group C sample selection. Samples for subgroups in group C shall be chosen at random from any inspection lot containing the intended package type and lead finish procured to the same specification which is submitted to and passes group A tests for quality conformance inspection. Testing of a subgroup using a single device type enclosed in the intended package type shall be considered as complying with the requirements for that subgroup.

4.4.3.4 Product acceptable for delivery. The product which may be accepted for delivery based a successful completion of group C testing shall be product of the particular package type with lot inspection identification codes of the 36 successive weeks beginning with the inspection lot identification code of the successful group C sample (i.e. the inspection lot identification codes accepted for delivery based on successful group C testing will include those sealed during the approximate 10 weeks during which the sample was being screened and tested plus those sealed in the 26 weeks following successful completion of group C testing making a total of 36 successive calendar weeks.) A manufacturer shall not accept product for delivery until after successful completion of the group C testing on product of the applicable package type with an inspection lot identification code less than 35 weeks prior to the inspection identification code of the product being considered for acceptance.

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4.5 Method of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows.

4.5.1 Pulse measurements. Conditions for pulse measurement shall be as specified in section 4 of MIL-STD-750.

4.5.2 Input capacitance. This test shall be conducted in accordance with method 3240 of MIL-STD-750, except the output capacitor shall be omitted.

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TABLE I. Group A inspection

Inspection 1/	MIL-STD-750		Symbol	Limit		Unit
	Method	Conditions		Min	Max	
Subgroup 1						
Visual and mechanical examination	2071	45 devices, c = 0				
Solderability 1/ 2/	2026	15 leads, c = 0				
Resistance to solvents 2/ 3/ 4/	1022	15 devices, c = 0				
Subgroup 2						
Breakdown voltage, collector to base	3001	Bias condition D; I_C = 10 uA dc	$V_{(BR)CBO}$	60		V dc
Breakdown voltage, emitter to base	3026	Bias condition D; I_E = 10 uA dc	$V_{(BR)EBO}$	5		V dc
Breakdown voltage, collector to emitter	3011	Bias condition D; I_C = 10 mA dc; pulsed (see 4.5.1)	$V_{(BR)CEO}$	60		V dc
Collector to base cutoff Current	3041	Bias condition D; V_{CE} = 50 V dc	I_{CES}		50	nA dc
Collector to base cutoff Current	3036	Bias condition D; V_{CE} = 50 V dc	I_{CBO1}		10	nA dc
Emitter to base cutoff current	3061	Bias condition D; V_{EB} = 4 V dc	I_{EBO}		50	nA dc
Forward-current transfer ratio	3076	V_{CE} = 10 V dc; I_C = 0.1 mA dc	h_{FE1}			
2N2906A, UA, UB				40		
2N2907A, UA, UB				75		
Forward-current transfer ratio	3076	V_{CE} = 10 V dc; I_C = 1.0 mA dc	h_{FE2}			
2N2906A, UA, UB				40	175	
2N2907A, UA, UB				100	450	
Forward-current transfer ratio	3076	V_{CE} = 10 V dc; I_C = 10 mA dc	h_{FE3}			
2N2906A, UA, UB				40		
2N2907A, UA, UB				100		

See footnotes at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limit		Unit
	Method	Conditions		Min	Max	
Subgroup 2 - Continued						
Forward-current transfer ratio	3076	V_{CE} = 10 V dc; I_C = 150 mA dc; pulsed (see 4.5.1)	h_{FE4}			
2N2906A, UA, UB				40	120	
2N2907A, UA, UB				100	300	
Forward-current transfer ratio	3076	V_{CE} = 10 V dc; I_C = 500 mA dc; pulsed see 4.5.1	h_{FE5}			
2N2906A, UA, UB				40		
2N2907A, UA, UB				50		
Collector-emitter saturation voltage	3071	I_C = 150 mA dc; I_B = 15 mA dc pulsed (see 4.5.1)	$V_{CE(sat)1}$			
2N2906A, UA, UB					0.4	V dc
2N2907A, UA, UB						
Collector-emitter saturation voltage	3071	I_C = 500 mA dc; I_B = 50 mA dc; pulsed (see 4.5.1)	$V_{CE(sat)2}$			
2N2906A, UA, UB					1.6	V dc
2N2907A, UA, UB						
Base-emitter saturation voltage	3066	Test condition A; I_C = 150 mA dc; I_B = 15 mAdc; pulsed (see 4.5.1)	$V_{BE(sat)1}$			
2N2906A, UA, UB				0.6	1.3	Vdc
2N2907A, UA, UB						
Base-emitter saturation voltage	3066	Test condition A; I_C = 500 mA dc; I_B = 50 mA dc; pulsed (see 4.5.1)	$V_{BE(sat)2}$			
2N2906A, UA, UB					2.6	V dc
2N2907A, UA, UB						
Subgroup 3						
High temperature operation		T_A = +150°C				

See footnotes at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limit		Unit
	Method	Conditions		Min	Max	
Subgroup 3 - Continued						
Collector to base cutoff current	3036	Bias condition D; V_{CB} = 60 V dc	I_{CBO2}		10	µA dc
Low temperature operation		T_A = -55°C				
Forward-current transfer ratio	3076	V_{CE} = 10 V dc; I_C = 10 mA dc	h_{FE6}			
2N2906A, UA, UB				20		
2N2907A, UA, UB				50		
Subgroup 4						
Small-signal short-circuit forward current transfer ratio	3206	V_{CE} = 10 V dc; I_C = 1 mA dc; f = 1 kHz	h_{fe}			
2N2906A, UA, UB				40		
2N2907A, UA, UB				100		
Magnitude of small-signal short-circuit forward current transfer ratio	3306	V_{CE} = 20 V dc; I_C = 20 mA dc; f = 100 MHz	$\lvert h_{fe} \rvert$	2.0		
Open circuit Output capacitance	3236	V_{CB} = 10 V dc; I_E = 0; 100 kHz ≤ f ≤ 1 MHz	C_{obo}		8	pF
Input capacitance (output open-circuited)	3240	V_{EB} = 2.0 V dc; I_C = 0; 100 kHz ≤ f ≤ 1 MHz	C_{ibo}		30	pF
Saturated turn-on time		(See figure 6)	t_{on}			
2N2906A, UA, UB					45	ns
2N2907A, UA, UB						
Saturated turn-off time		(See figure 7)	t_{off}			
2N2906A, UA, UB					300	ns
2N2907A, UA, UB						
Subgroups 5, 6, and 7						
Not required						
Subgroup 8 4/						
Decap internal visual (design verification)	2075	1 device, c = 0				

1/ For sampling plan see MIL-S-19500.
2/ Separate samples may be used.
3/ Not required for laser marked devices.
4/ Not required for JANS devices.

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TABLE II. Groups A, B, and C electrical end-point inspection measurements.

Step	Inspection	MIL-STD-750		Symbol	Limit		Unit
		Method	Conditions		Min	Max	
1.	Collector-base cutoff current	3036	Bias condition D;	I_{CBO1}			
	2N2906A,UA,UB		V_{CB} = 50 V dc			20	nA dc
	2N2907A,UA,UB					10	nA dc
2.	Collector - base cutoff current	3036	Bias condition D;	I_{CBO2}			
	2N2906A,UA,UB		V_{CB} = 50 V dc			40	nA dc
	2N2907A,UA,UB					20	nA dc
3.	Emitter-base cutoff current	3061	Bias condition D; V_{EB} = 3.5 V dc	I_{EBO}		10	nA dc
4.	Collector to emitter voltage (saturated)	3071	I_C = 150 mA dc; I_B = 15 mA dc pulsed (see 4.5.1)	$V_{CE(sat)1}$			
	2N2906A,UA,UB					.4	V dc
	2N2907A,UA,UB						
5.	Base-emitter voltage (saturated)	3066	Test condition A V_{CE} = 10 V dc; I_C = 1.0 mA dc pulsed (see 4.5.1)	$V_{BE(sat)1}$			
	2N2906A,UA,UB				0.6	1.3	V dc
	2N2907A,UA,UB						
6.	Forward current transfer ratio	3076	V_{CE} = 10 V dc; I_C = 1.0 mA dc	h_{FE2}			
	2N2906A,UA,UB				40	175	
	2N2907A,UA,UB				100	450	
7.	Collector-base cutoff current	3036	Bias condition D V_{CB} = 60 V dc	ΔI_{CBO1} 1/	100% of initial value or 8 nA dc, whichever is greater.		
	2N2906A,UA,UB						
	2N2907A,UA,UB						
8.	Forward current transfer ratio	3076	V_{CE} = 10 V dc; I_C = 1.0 mA dc; pulsed see 4.5.2	Δh_{FE2} 1/	±25% change from initial reading.		

1/ Devices which exceed the group A limits for this test shall not be accepted.

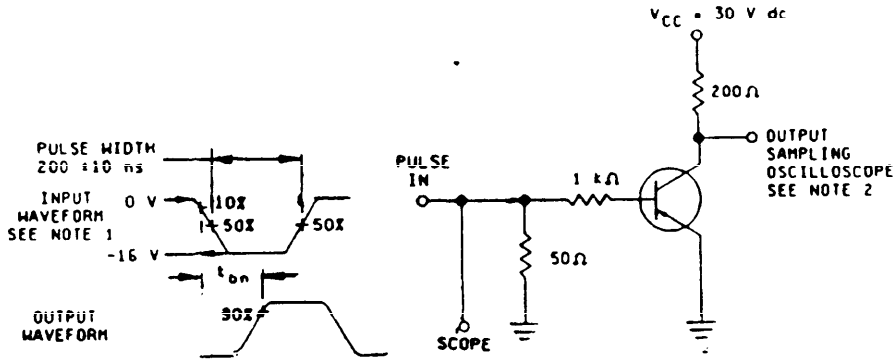
2/ The electrical measurements for table IVa (JANS) of MIL-S-19500 are as follows:
- a. Subgroup 3, see table II herein, steps 1, 2, 3, 4, 5, and 6.
- b. Subgroups 4 and 5, see table II herein, steps 2, 3, 4, 5, and 8.

3/ The electrical measurements for group B, (4.4.2.2 herein) are as follows:
- a. Subgroups 1 and 2, see table II herein, steps 1, 6, 7, and 8.

4/ The electrical measurements for table V (JANS) of MIL-S-19500 are as follows:
- a Subgroup 2 and 3, see table II herein, steps 1, 2, 3, 4, 5, and 6 (for JANS).
- b. Subgroup 6, see table II herein, steps 2, 3, 4, 5, 6, 7, and 8 (for JANS).

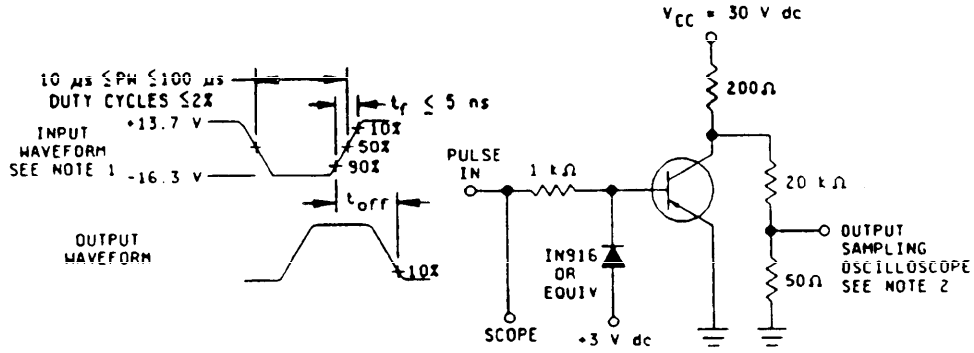
5/ The electrical measurements for table V (JAN, JANTX, and JANTXV) of MIL-S-19500 are as follows:
- a. Subgroups 2, 3, and 6, see table II herein, steps 1 and 6.

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NOTES:
1. The rise time (t_r) of the applied pulse shall be ≤ 2.0 ns, duty cycle ≤ 2 percent, and the generator source impedance shall be 50Ω.
2. Sampling oscilloscope: Z_{in} ≥ 100 kΩ, C_{in} ≤ 12 pF, rise time ≤ 5 ns.

FIGURE 6. Saturated turn-on switching time test circuit.



NOTES:
1. The rise time (t_r) of the applied pulse shall be ≤ 2.0 ns, duty cycle ≤ 2 percent, and the generator source impedance shall be 50Ω.
2. Sampling oscilloscope: Z_{in} ≥ 100 kΩ, C_{in} ≤ 12 pF, rise time ≤ 5 ns.

FIGURE 7. Saturated turn-off switching time test circuit.

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5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-S-19500.

6. NOTES
(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Notes. The notes specified in MIL-S-19500 are applicable to this specification.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Lead formation and finish may be specified (see 3.3.1).

b. Type designation and product assurance level.

c. For die acquisition, the JANHC or JANKC letter version shall be specified (see figure 5).

d. Surface mount designation if applicable.

6.3 Supersession information. Devices covered by this specification supersede the manufacturers' and users' Part or Identifying Number (PIN). The term Part or Identifying Number (PIN) is equivalent to the term part number which was previously used in this specification. This information in no way implies that manufacturers' PIN's are suitable as a substitute for the military PIN.

6.4 Suppliers of JANHC and JANKC die. The qualified JANHC and JANKC suppliers with the applicable letter version (example JANHCA2N2906A) will be identified on the QPL.

JANC ordering information		
	Manufacturer	
PIN	04713	
2N2906A	JANHCA2N2906A	
2N2907A	JANHCA2N2907A	
2N2906A	JANKCA2N2906A	
2N2907A	JANKCA2N2907A	

6.5 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 17
NASA - NA

Review activities:
Army - AR, MI, SM
Navy - AS, CG, MC, SH
Air Force - 13, 15, 19, 85, 99
DLA - ES

Preparing activity:
Army - ER

Agent:
DLA - ES

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INSTRUCTIONS

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-S-19500/291E	2. DOCUMENT DATE (YYMMDD) 28 July 1994 940728

3. DOCUMENT TITLE SEMICONDUCTOR DEVICE, TRANSISTOR, PNP, SILICON, SWITCHING, TYPES 2N2906A, 2N2907A, 2N2906AUA 2N2907AUA, 2N2906AUB, AND 2N2907AUB JAN, JANTX, JANTXV, JANS, JANHC, AND JANKC

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME Patrick Layden	b. TELEPHONE (Include Area Code) (1) Commercial 908 544-2414	(2) AUTOVON 995-2414
c. ADDRESS (Include Zip Code) US Army Research Laboratory and Power Sources Directorate ATTN: AMSRL-EP-RD Ft. Monmouth, NJ 07703-5601	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756 2340 AUTOVON 289-2340	

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